

Competitive Effects of State Antitrust Laws: Evidence from the Progressive Era

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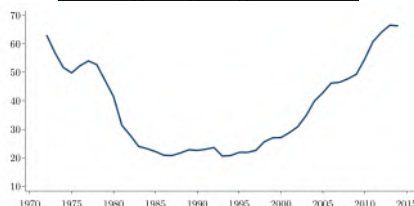
September 30, 2025

Rising Concentration in the U.S. Economy Since the 1980s

Average Markups



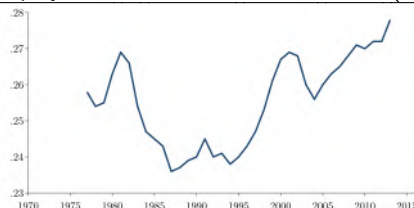
Median Firm Size (\$Millions)



HHI



Employment in 10,000+ Worker Firms (%)



Sources: De Loecker, Eeckhout, and Unger (2020) and Grullon, Larkin, and Michaely (2019).

Calls for Strengthening U.S. Antitrust Policy

Market Power Has Grown and Antitrust Needs Strengthening, Despite What Shapiro & Yurukoglu and Miller Suggest

Jonathan Baker and Fiona Scott Morton in *ProMarket*, October 1, 2024

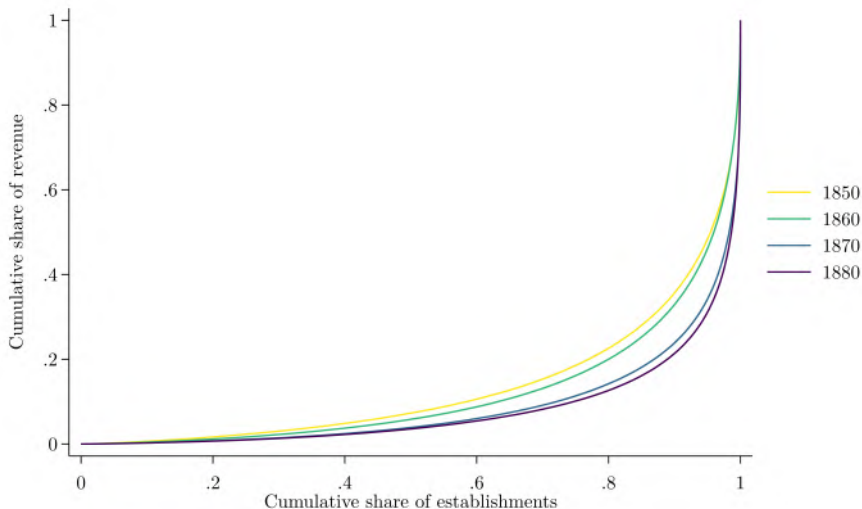
Hawley Op-Ed: Break Up Big Tech Behemoths to Reclaim American Freedom

Josh Hawley in *The New York Post*, May 1, 2025

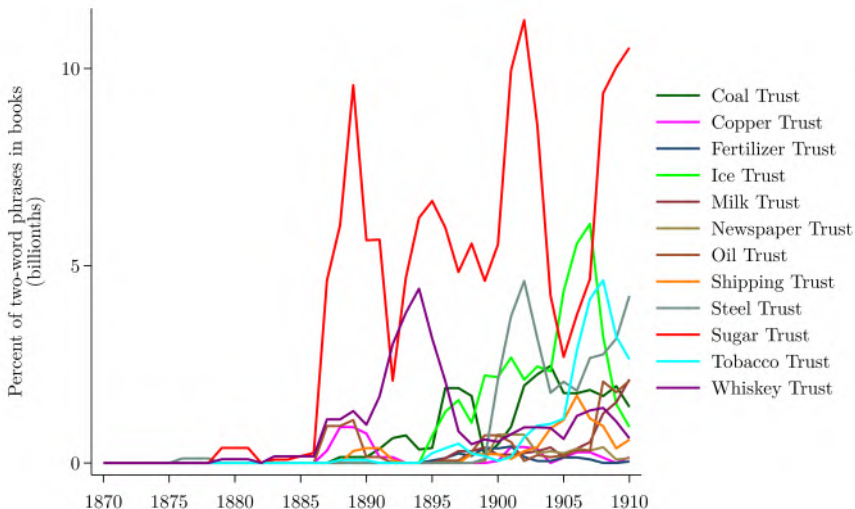
Business Community Unites in Call for Increased Antitrust Enforcement Funding

National Grocers Association, June 4, 2025

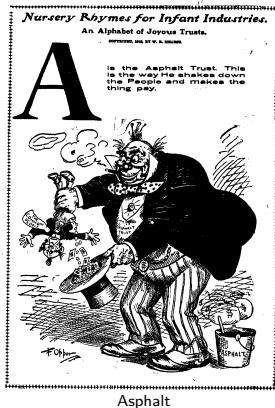
Concentration of Revenue in Manufacturing, 1850-1880



Mentions of Well-Known Trusts in *Google Books*



Nursery Rhymes for Infant Industries



Beef



Coal



Dough



Electricity



Fruit



Glass



Hydr. Brick



Ice



Jag



Kodak



Lumber



Matches



Nat. Biscuit



Oil



Pianos



Quinine



Railroads



Shipping



Theater



U.S. Rubber



Varnish



Wallpaper



Xerxes



Yarn



Zephyr

LAWS WHICH DEAL WITH "TRUST" PROBLEM

All Over the Country Legislators Are Planning to Combat the Nineteenth Century Business Octopi.

ANALYSIS OF THE STATUTES OF VARIOUS STATES.

Leading Features of the Measures That Are Now in Force in Sixteen Commonwealths.

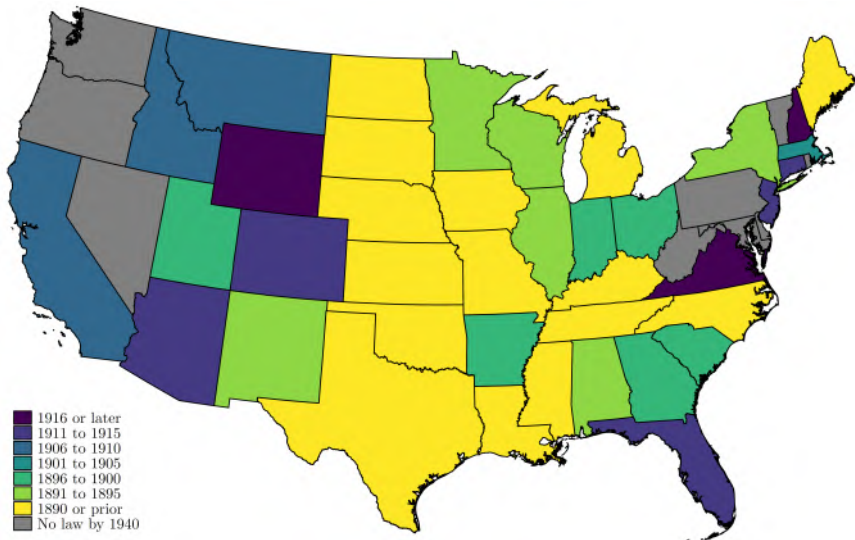
TWENTY-ONE ARE NOW WITHOUT SUCH ENACTMENTS.

Illinois, Kansas, Kentucky, Mississippi, Missouri, and Nebraska Prohibit the Combinations.

LEGISLATION NOW PENDING IN SEVEN GENERAL ASSEMBLIES.

Chicago Daily Tribune
February 20, 1897

Map of State Antitrust Law Adoption, 1888-1940



State Antitrust Laws Were Portrayed as Significant



Los Angeles Times
May 16, 1907

- The passage and enforcement of state antitrust laws frequently generated substantial media coverage
- State attorneys general brought numerous cases, including actions against large interstate firms
- In some cases, offenders faced severe penalties

Research Question

- How did antitrust statutes affect the economies of the states that enacted them?
 - I examine effects on the number of establishments, average profits, employment, wages, and the labor share in the manufacturing sector
 - I also examine effects on patenting behavior as a proxy for innovation

Preview of Results

- I find that state antitrust laws:
 - Increased the number of manufacturing establishments by 10%, which likely reflects growth by existing firms rather than entry
 - Decreased the labor share by about 4% in the manufacturing sector overall and 7% in trust-affiliated industries
 - Had no discernible effect on manufacturing establishments' profits or labor demand
- I find no evidence that more stringent laws produced stronger procompetitive effects
- I also document an increase in patents granted to firms of about 7.5% and an increase in patents granted to individuals of about 5%

Contribution

Prior Literature and Contribution

- Effects of U.S. antitrust policy on competition:

Babina et al. 2023; Baker, Frydman, and Hilt 2023; Binder 1988; Bittlingmayer 1992; Bittlingmayer 1993; Bosch and Eckard 1991; Burns 1977; Gil 2015; Mullin, Mullin, and Mullin 1995; Mullin and Snyder 2021; Peltzman 2014; Prager 1992; Shepherd 1982; Sproul 1993; Stigler 1966; Young and Shughart 2010

- Effects of U.S. antitrust policy on innovation:

Hashmi 2013; Kang 2025; Kwon and Marco 2021; Thatchenkery and Katila 2023; Watzinger et al. 2020; Watzinger and Schnitzer 2022

Contribution:

- Estimate market-wide effects of instituting an antitrust regime—beyond single cases—capturing potential deterrence effects
- Measure impacts across the manufacturing sector, not select industries
- Track outcomes over nearly a century to assess persistence and timing

Prior Literature and Contribution

- Collusion and firm behavior in the historical United States:

Alexander 1994; Chicu, Vickers, and Ziebarth 2013; Gross 2020; Huang 2025; Porter 1983; Syverson 2024; Vickers and Ziebarth 2014

- Significance of early state competition policy:

Clay and Troesken 2002; Lamoreaux 2023; Lamoreaux and Phillips-Sawyer 2021; May 1987; Nolette 2012; Phillips-Sawyer 2018; Pratt 1980; Troesken 1995; Troesken 1998; Troesken 2000

Contribution:

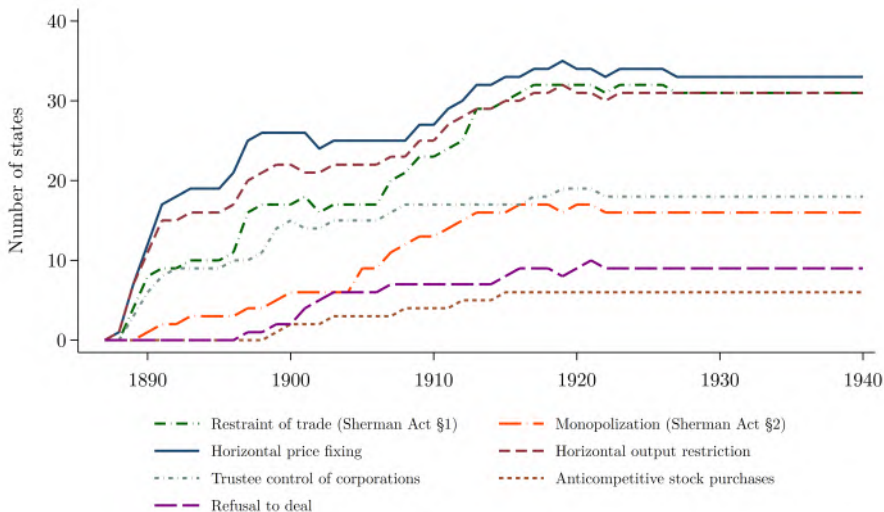
- Use quasi-experimental methods to quantify the effects of state laws
- Gauge “success” via economic indicators, not legal wins
- Trace the evolution of state antitrust statutes over time

Background

The Origins of State Antitrust Laws

- The origins of antitrust have received significant scholarly attention as courts rely on legislative intent to guide modern judicial application
- The standard explanation is that “agrarian discontent”—lobbying against railroads and meatpackers by farmers—was a key driver of early antitrust legislation
- Ongoing work (Mayo and Schaller 2025) seeks to understand the drivers of antitrust legislation using quantitative methods that:
 - Compare the explanatory power of competing explanations for the adoption of state antitrust laws
 - Identify themes in large-scale newspaper data on monopolies

Number of States with Statutory Bans, by Illegal Behavior



What Does a Typical State Antitrust Law Do?

- Define illegal activities

Crimes vs. Torts

- Establish damages to be awarded to injured parties for civil infractions

Damage levels

- Establish fines and/or prison sentences for criminal infractions

Fines

Prison sentences

Daily offenses

- Empower the state to revoke the charters of offending corporations

Revocations

- Assign enforcement to one or more state officials

Enforcement duties

Antitrust Federalism

- State-level action laid the groundwork for reform at the federal level, beginning with the Sherman Antitrust Act of 1890
- The legislative record shows Congress intended federal antitrust law to supplement, not supplant, state enforcement, and created an additional forum for *interstate* restraints of trade (Hovenkamp 1983)
- Courts typically focused on whether *restraints* crossed state lines—not whether their effects did—leaving room for state enforcement even when interstate commerce was affected (Brinkerhoff 2017) Examples
- Antitrust enforcement was mostly state-led until the New Deal era's expansion of federal commerce power (Columbia Law Review 1961)
- In its early years, federal antitrust enforcement was notoriously weak

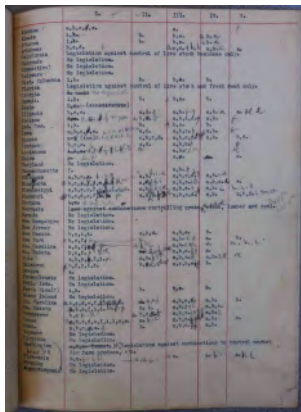
Expected Effects of Antitrust and Increased Competition

Outcome	Expected Effect	Mechanism
Number of firms	↑	Reduced barriers to entry
Profits	↓	Collusive structures are replaced by more competitive ones
Employment	↑	Output expands as collusion relaxes
Wages	↑ or ↓	Expansion of output can increase labor demand, but thinner margins create downward pressure on wages
Labor share	↑	Increased employment and thinner margins increase the fraction of income paid as wages
Innovation	↑ or ↓	Rivalry spurs innovation up to a point, but large firms have the resources to fund R&D

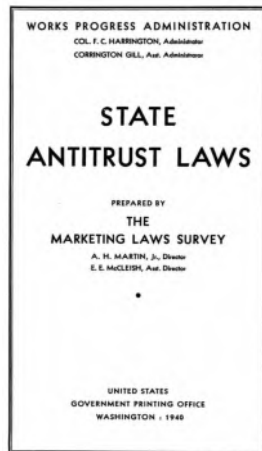
Data

Building a Dataset on State Antitrust Statutes

- I searched state session laws and codes for keywords typically present in antitrust statutes
- I compared my dataset to existing compilations
- I identified cases overturning laws in part or in full
- I coded the content of all statutes

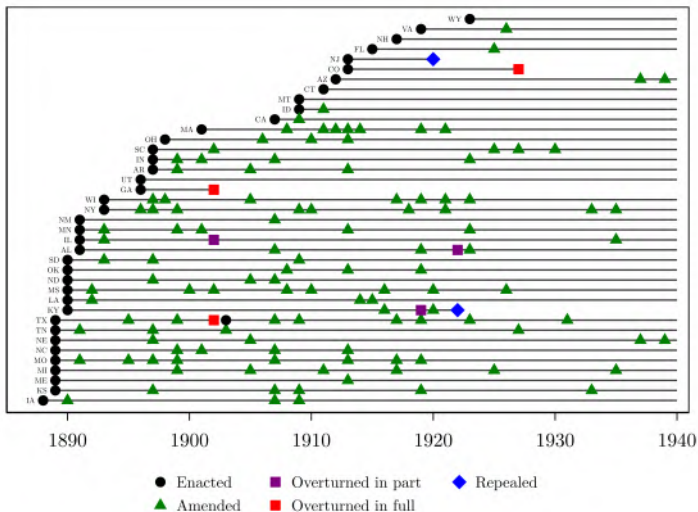


1903 internal memo,
Bureau of Corporations



1940 published report,
Works Progress Administration

State Antitrust Statutes, 1888–1940



Census of Manufactures Data, 1850–1939

STATISTICS OF MANUFACTURES.

39

TABLE IV.—MANUFACTURES IN EACH STATE AND TERRITORY, 1880.

By value added, by principal industries—Continued.

Manufacturing industries.	No. of establishments.	Value added, \$.	Value added, \$.		Value added, \$.	Value added, \$.	Value added, \$.	Value added, \$.
			By product.	By material.				
Food and kindred products.	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000
Textile millinery and millinery goods.	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000
Woolen and worsted manufactures.	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000
Cotton textile manufactures.	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000
Other textile manufactures.	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000
Other manufactures.	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000
Total.	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000	1,100,000

Example of a state-by-industry tabulation from the 1880 census of manufactures

- I digitized state-by-industry tabulations from the 1850–1992 censuses of manufactures in related co-authored work (Barkai, Karger, and Schaller 2025)
- I use data for 1850–1939
- The data span each of the 20 manufacturing censuses that took place between 1850 and 1939 Data coverage
- I map industries to their corresponding four-digit SIC codes

The Truth About the Trusts, 1904

The Truth About The Trusts

A Description and Analysis of the
American Trust Movement

By JOHN MOODY

EDITOR, "Moody's Manual of Corporation Securities"



MOODY PUBLISHING COMPANY

35 Nassau Street, New York

CHICAGO OFFICE: 79 Dearborn Street

- In the absence of firm-level data, *The Truth About the Trusts* provides detailed descriptions on 445 trusts in the United States at the height of the merger wave (Moody 1904)
- I infer each trust's primary industry and assign a four-digit SIC code
- *The Truth About the Trusts* draws on *Moody's Manual of Corporation Securities*, the *Commercial & Financial Chronicle*, the *Boston News Bureau*, the *Wall Street Journal*, and other sources

Comprehensive Universe of U.S. Patents, 1850–1940

- I use data from the Comprehensive Universe of U.S. Patents (CUSP) to gauge the extent to which state antitrust laws promoted innovation (Berkes 2018)
- CUSP covers both assignee (firm-sponsored) patents and inventor (individual) patents
- CUSP includes all granted U.S. patents and published applications, covering a wide range of technologies
- The data include detailed bibliographic information, including inventors, assignees, filing dates, classification codes, and citations

Historical Newspaper Articles, 1850–1940

- I use data from the Library of Congress's *Chronicling America* project and the ProQuest Historical Newspapers database
- To identify discussion of state antitrust laws, I gather articles that:
 - 1 Contain either antitrust or anti-trust
 - 2 Contain a state name or the title of a named state antitrust law (e.g., cartwright in reference to California's Cartwright Act)
 - 3 Were published between 1850 and 1940
- I remove articles containing words and names (such as the names of U.S. attorneys general) that likely reflected federal antitrust actions
- This search ultimately generates a collection of 118,560 articles

Examples of Newspaper Articles in My Dataset

**STANDARD OIL
FOUND GUILTY**

Combine Convicted of Violating the Anti-Trust Law of Ohio.

Special Dispatch to the "Chronicle."
FINDLAY (O.), October 19.—By the

**CASH REGISTER CO. GIVEN
\$10,000 FINE IN MICHIGAN.**

Anti-Trust Law Charge by Rival Corporation Brings Conviction of Dayton, O., Firm.

Lansing, Mich., July 25.—The state Supreme Court today found the National Cash Register company of Dayton, O., guilty of unlawful restraint of trade and imposed a fine of \$10,000.

The company is restrained from doing business in Michigan until the fine is paid.

**BREWERS CONFER WITH
TEXAS ATTORNEY GENERAL.**

Willing to Confess Judgment in Anti-Trust Suits—Pabst Not Represented.

Special to The New York Times.

AUSTIN, Texas, Nov. 16.—Attorneys representing the Anheuser-Busch and the Lemps Brewing Companies of St. Louis, the American Brewing Company and the Houston Ice and Brewing Company of Houston, the San Antonio Brewing Company and the Lone Star Brewing Company of San Antonio, and the Galveston Brewing Company of Galveston, to-day held a

CROOKED LEGISLATORS.

It Took \$50,000 to Smother an Anti-Trust Bill in Illinois.

CHICAGO, May 24.—A morning paper publishes a dispatch from Springfield, giving an alleged expose of lobby work in the State Legislature. Some time ago an anti-trust bill was passed by the House with very little opposition. It is now in the Senate and so placed that it will require a two-thirds majority to call it up. It is not believed that this majority can be secured, and that the bill will die with the adjournment of the Legislature without having been acted on. The correspondent, it seems, went to Jacob Bunn, president of the Illinois Watch Company, under the guise of an emissary from President Avery, of the Elgin Watch Company, to learn the status of affairs.

INSURANCE IN KENTUCKY.
Indictments Under Anti-Trust Laws Against New York Companies.

CINCINNATI, Ohio, March 22.—The Grand Jury of Kenton County, at Covington, Ky., returned indictments at noon to-day against the Home Life Insurance Company of New York, and forty-two other life and fire companies, for "banding together and fixing, and maintaining a higher rate of premium than would otherwise prevail." The indictment is found under the anti-trust laws of the Commonwealth, and this action of the Grand Jury is

Trusts Must Go.

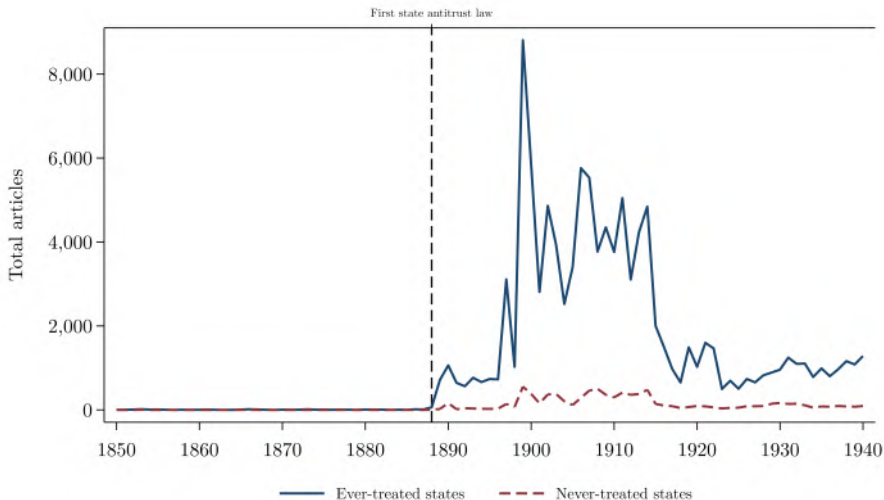
St. Louis, Nov. 23.—Two charters of about seven hundred trusts doing business in Missouri, and who have failed to comply with the anti-trust law, were revoked to-day. It is claimed that the law is unconstitutional, and the trust companies will fight the matter in the courts.

ICE TRUST PROSECUTED.

Accused of Violating Anti-Trust Laws in New York.

New York, Oct. 4.—The American Ice Company, through its counsel, to-day entered pleas of not guilty to four indictments charging the company with operating as a monopoly in restraint of trade. The arraignment was before Judge Wheeler of the state

Newspaper Articles Mentioning State Antitrust Laws



Control Variables

- Throughout the analysis, I control for the population, literacy rate, median occupational score, and personal income
- I use full-count population censuses to compute controls most controls (Ruggles et al. 2021)
 - I linearly interpolate missing data (e.g., for 1890)
- I create a composite series on state personal income drawn from Bureau of Economic Analysis (2025), Easterlin (1960), Easterlin (1957), and Klein (2009) Personal income series

Identification Strategy

Identification Strategy

- Because adoption of state antitrust laws was staggered and treatment effects may have varied over time, I employ a “stacked” difference-in-differences design (Baker, Larcker, and Wang 2022; Cengiz et al. 2019; Deshpande and Li 2019; Gormley and Matsa 2011)

$$y_{aist} = \beta_0 + \beta_1(\text{statelaw}_s \times \text{post}_{st}) + \Gamma X_{st} + \delta_{as} + \delta_{at} + \delta_{is} + \delta_{it} + \varepsilon_{aist} \quad (1)$$

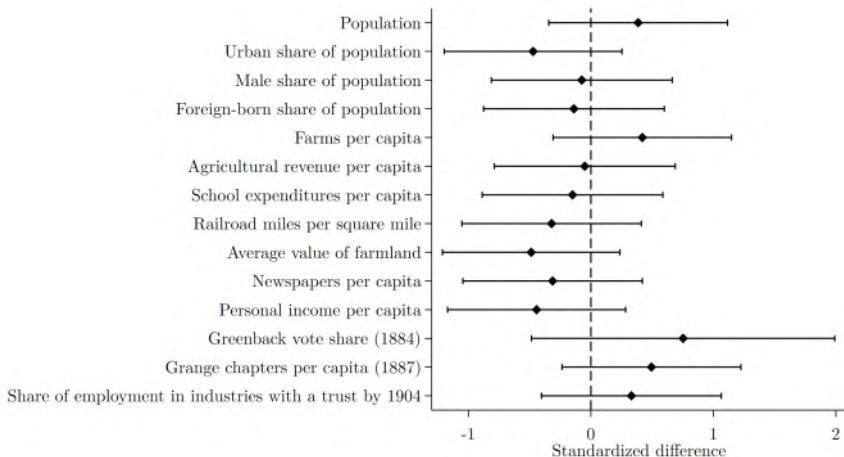
where where s , t , i , and a denote states, years, industries, and stacks (i.e., year-of-enactment cohorts), respectively

- Critically, this approach includes both state-by-industry fixed effects δ_{is} and industry-by-year fixed effects δ_{it}

Identification Strategy

- I exclude states whose antitrust laws were overturned or repealed throughout the analysis
- I cluster standard errors at the state level throughout the analysis
- I weight observations by revenue and by the inverse frequency of the census of manufactures (i.e., decennial=1, quinquennial=0.5, and biennial=0.2)
- Identifying assumption: In the absence of treatment, outcomes in treated and control states would have evolved similarly

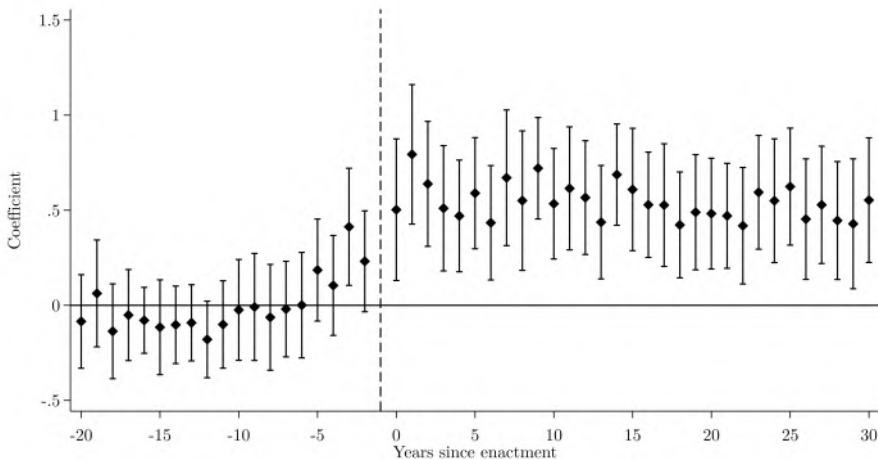
Ever-Treated vs. Never-Treated States in 1880



Results

Event Study Results: Log Number of Newspaper Mentions

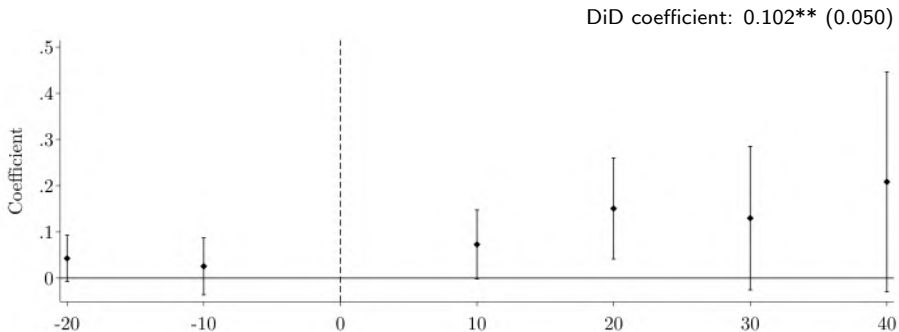
DiD coefficient: 0.691*** (0.241)



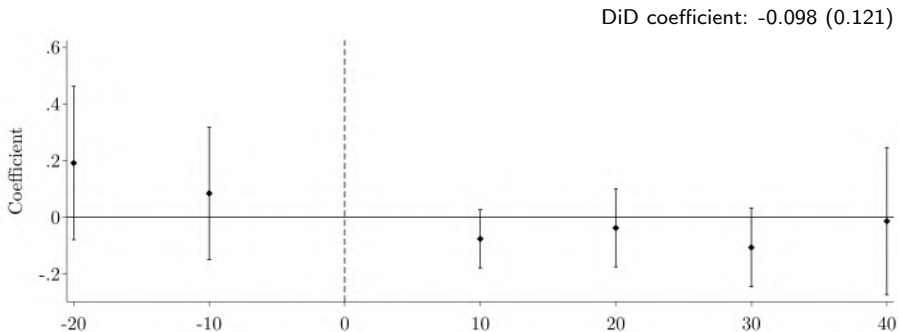
Results

Manufacturing Outcomes

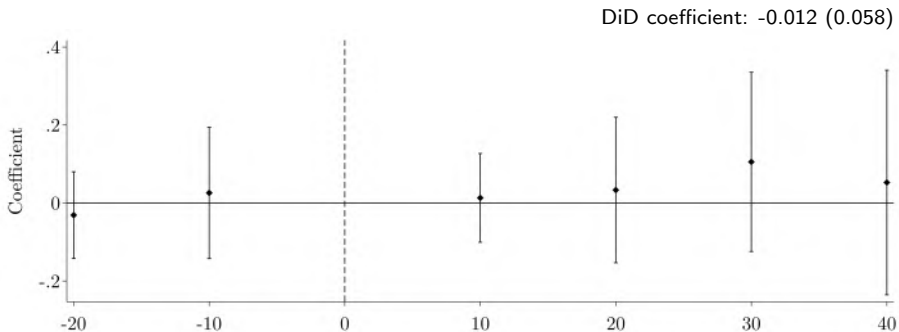
Event Study Results: Log Number of Establishments



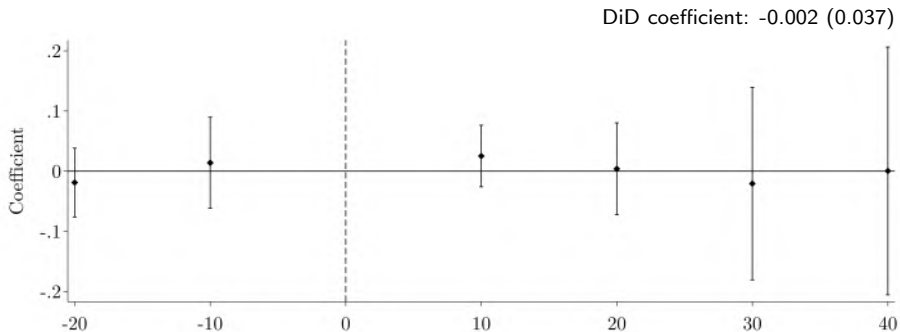
Event Study Results: Log Average Number of Proprietors



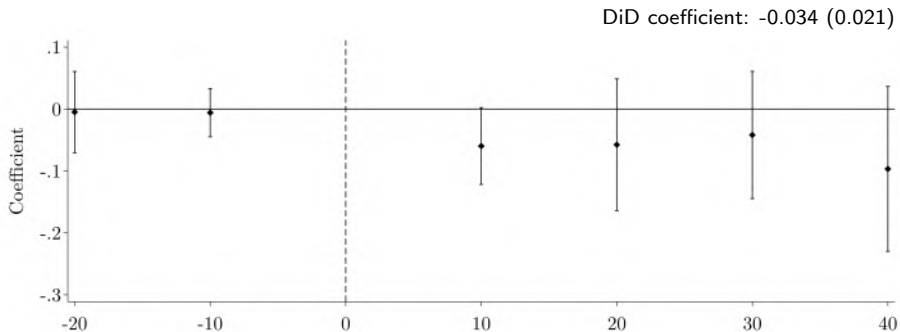
Event Study Results: Log Average Profit



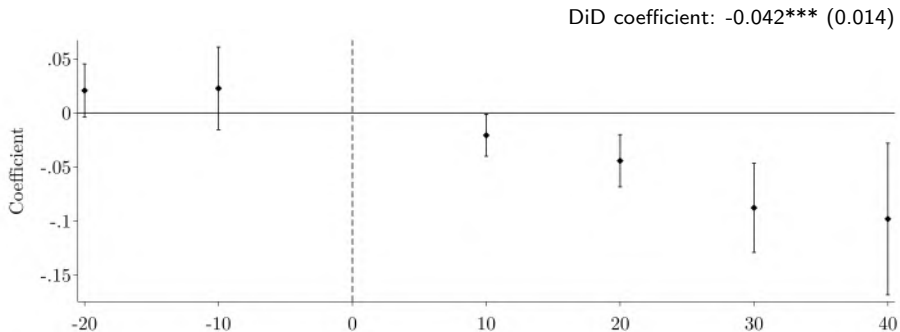
Event Study Results: Log Average Employment



Event Study Results: Log Average Wage

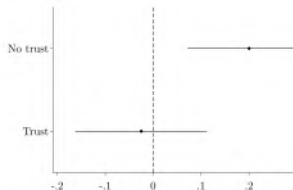


Event Study Results: Log Labor Share

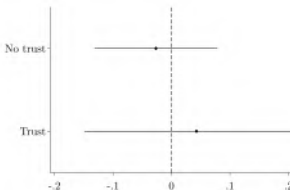


Manufacturing DiD Results, by Trust Status

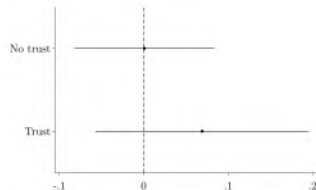
Number of Establishments



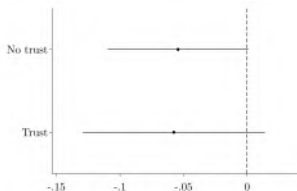
Average Profit



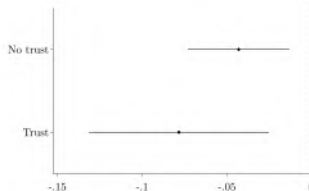
Average Employment



Average Wage



Labor Share



Results

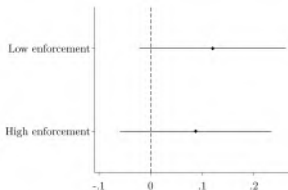
Enforcement Intensity

Proxying States' Enforcement Intensity

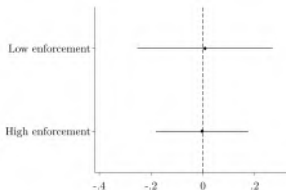
- Did differences in enforcement shape the effects of antitrust statutes?
- Problem: Records of cases brought by attorneys general and district attorneys are not consistently available; nor are data on private suits
- Solution: Use contemporaneous newspaper discussions to classify states as high- or low-enforcement and estimate treatment effects for each group relative to untreated states
- States are classified as high-enforcement based on the share of treated years with above-median newspaper mentions per capita

Manufacturing DiD Results, by Enforcement Level

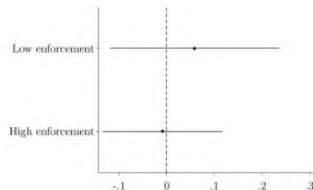
Number of Establishments



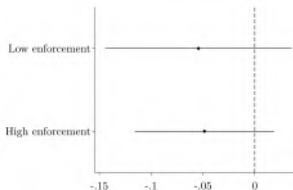
Average Profit



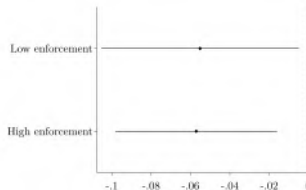
Average Employment



Average Wage



Labor Share



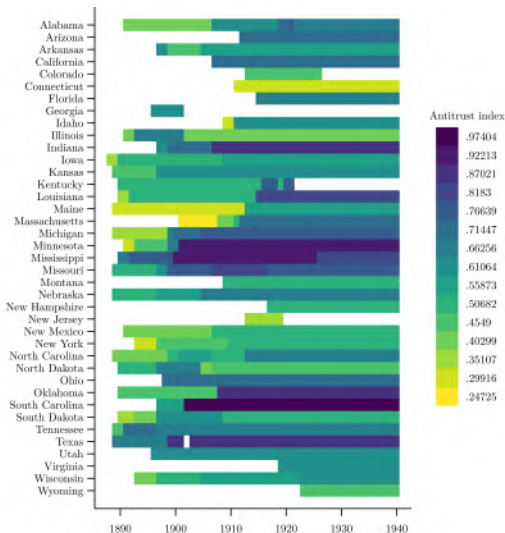
Results

Statutory Stringency

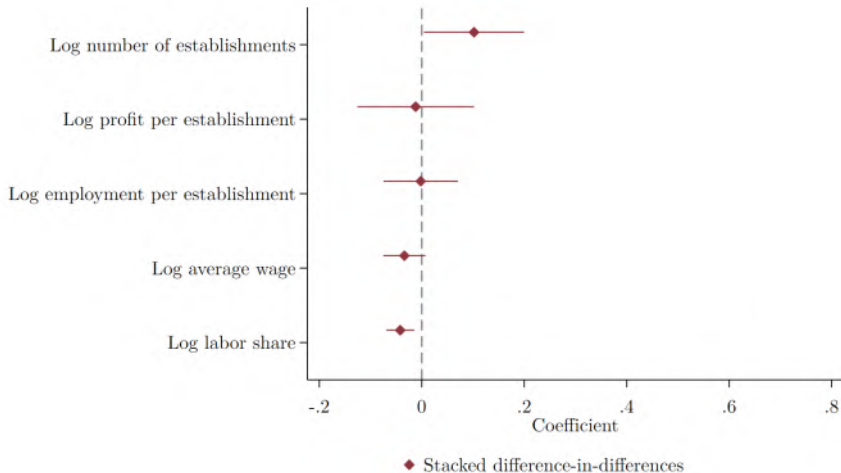
Developing an Antitrust Law Index

- To summarize the features of state antitrust laws with a single, continuous measure of stringency, I constructed an antitrust law index using Principal Component Analysis (PCA)
- To do this, I coded 19 numeric measures of statutory severity for each state and year in which an antitrust law was in force Variables with loadings
- I then standardized these variables and used PCA to extract the first principal component, which is the linear combination that maximizes the proportion of the total variance explained
- Final index: untreated states are $= 0$ and treated states are $\in [0.22, 1]$

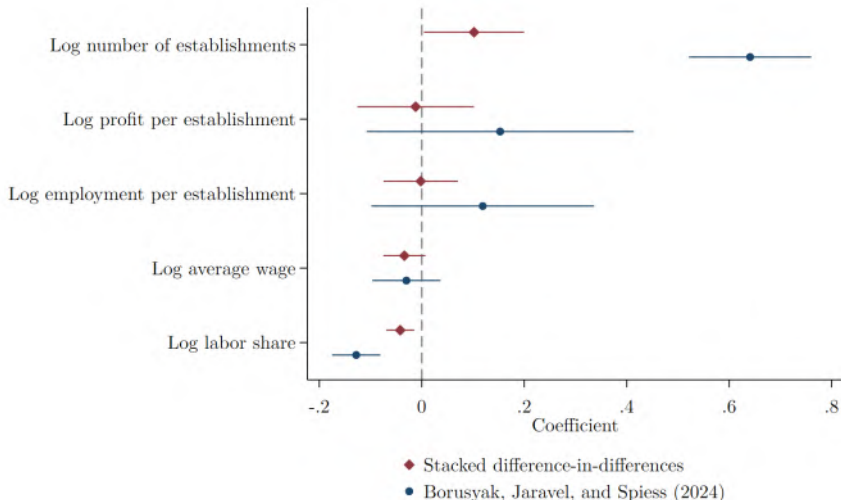
Antitrust Law Index by State, 1888–1940



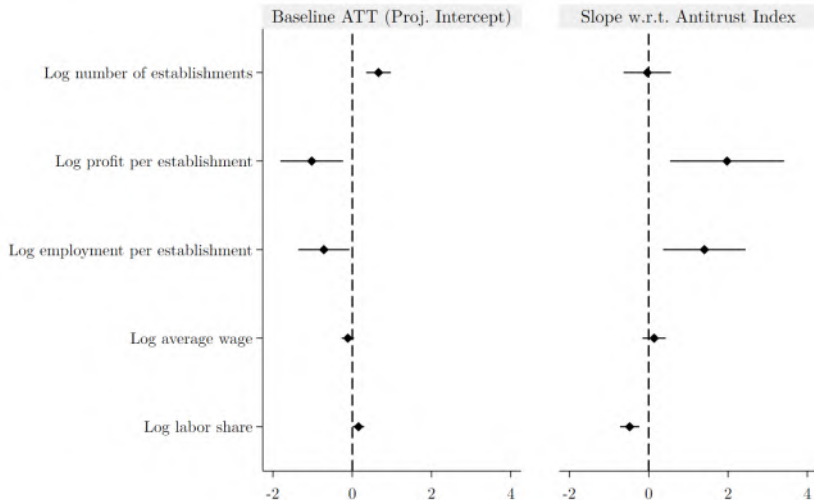
DiD Results for Manufacturing Outcomes



DiD Results for Manufacturing Outcomes



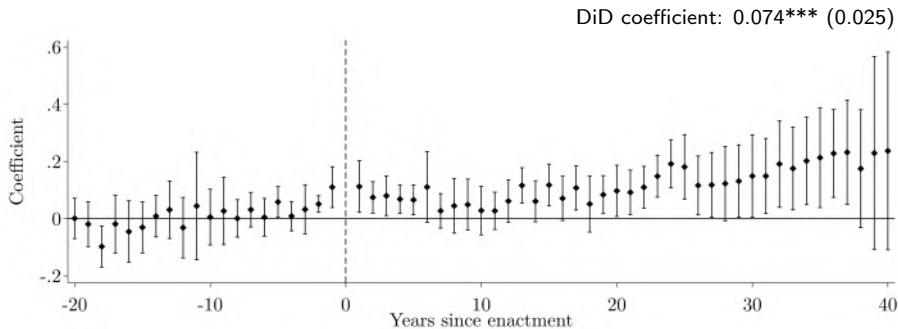
Continuous DiD Results for Manufacturing Outcomes



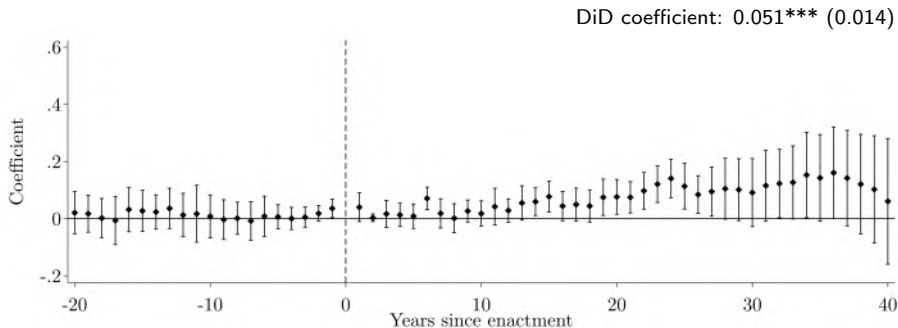
Results

Patenting Outcomes

Event Study Results: Log Number of Assignee Patents



Event Study Results: Log Number of Inventor Patents



Results

Robustness Checks and Additional Analyses

Robustness Checks

- Alternative weighting schemes Results
- Control for the share of agricultural employment Results
- Control for anti-monopoly constitutional provisions Results Provisions
- Labor share as total wages over value added Results

Additional Analyses

- Manufacturing productivity [Residual with Capital](#) [LP with Capital](#) [LP with Establishments](#)
- Manufacturing results by industry [Results](#)
- Repealed or struck-down laws [Results](#)
- Average daily stock returns around enactment [Results](#)
- Incorporations by trusts [Results](#) [The Great Merger Movement](#)
- Food retail prices [Results](#)

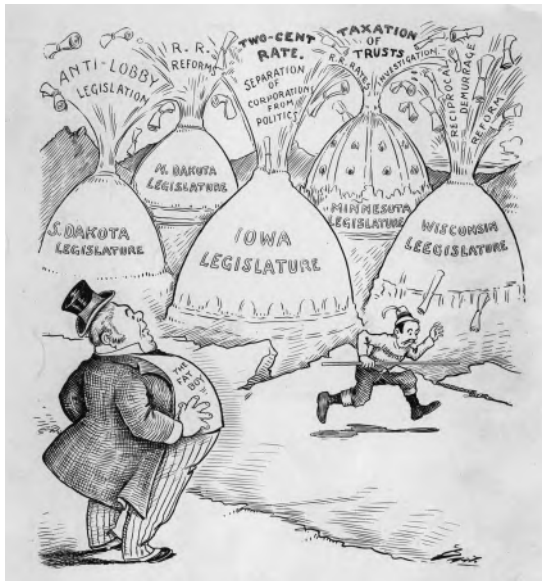
Conclusion

Actual Effects of State Antitrust Laws

Outcome	Expected effect	Actual effect
Number of firms	↑	—
Profits	↓	—
Employment	↑	—
Wages	↑ or ↓	—
Labor share	↑	↓
Innovation	↑ or ↓	↑

Conclusion

- I find that state antitrust statutes entrenched rather than curtailed market power
 - Establishments increased 10%, likely driven by incumbent expansion rather than new entry
 - The labor share declined by 7% in trust-affiliated industries.
 - Profits and employment remained unchanged
 - Patenting increased, suggesting that “bigness” fuels innovation
- Broader lesson: legal reforms alone are insufficient to shift markets towards competitive equilibria

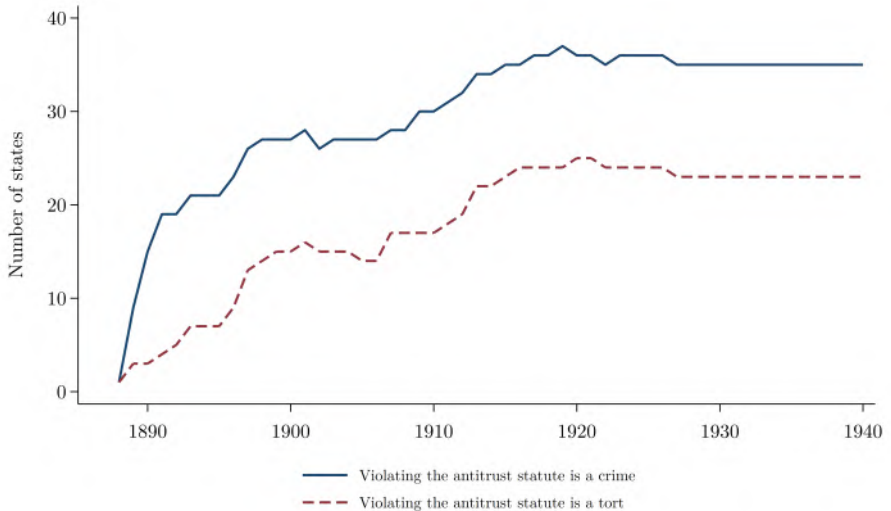


The Minneapolis Journal
January 31, 1907

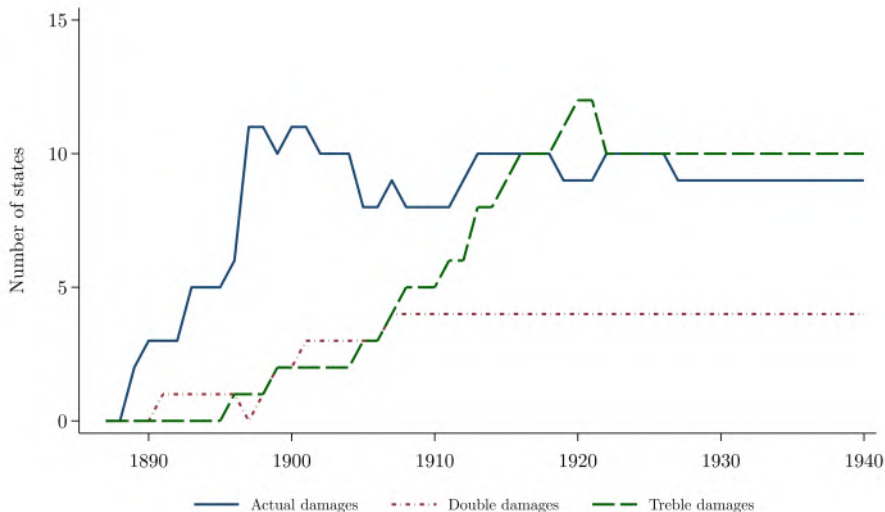
*Puck*

September 7, 1904

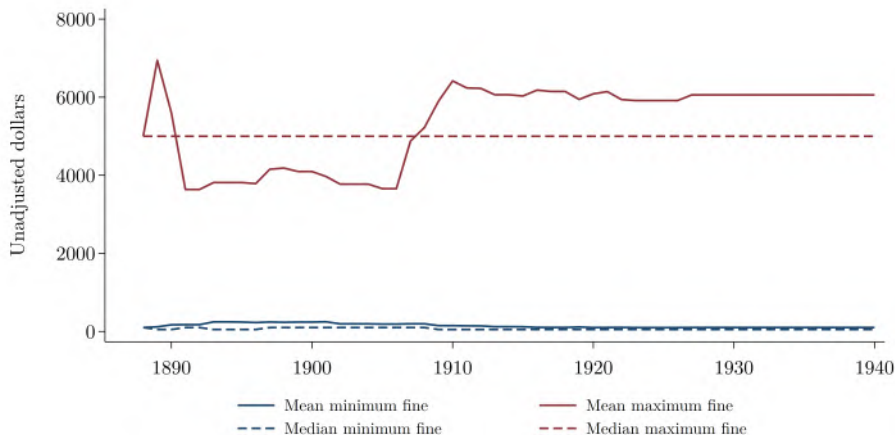
Number of States Declaring Violations Crimes and/or Torts



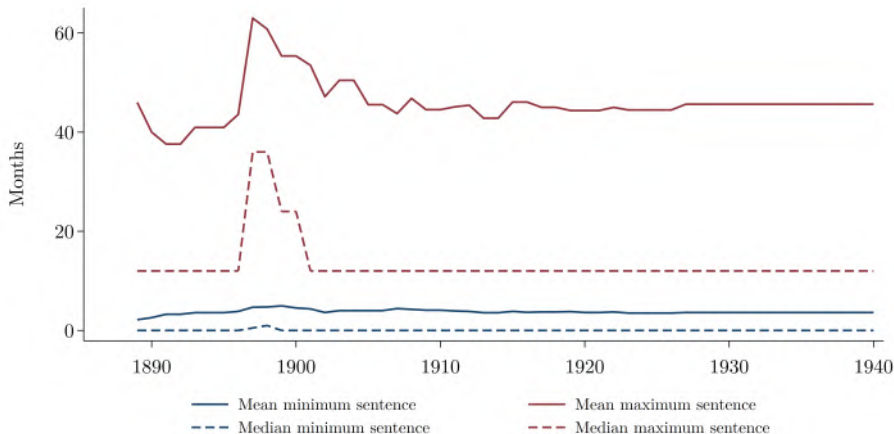
Number of States Allowing Civil Damages, by Amount



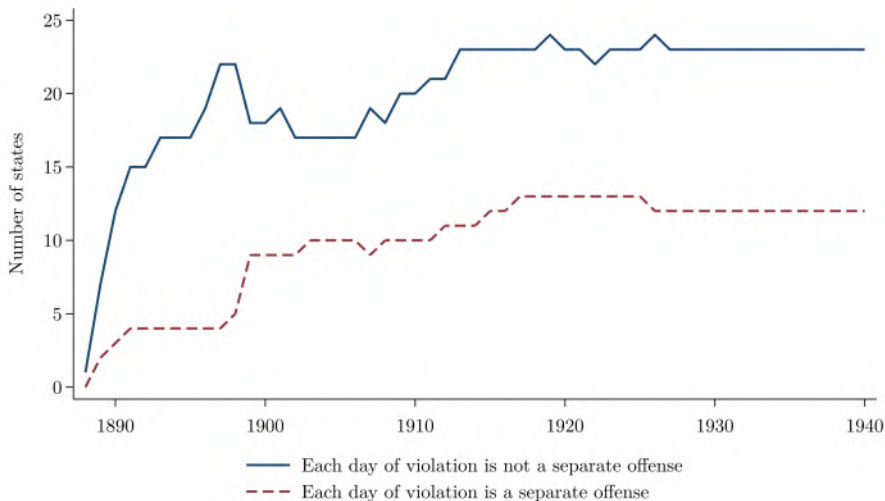
Avg. Statutory Minimum and Maximum Fines



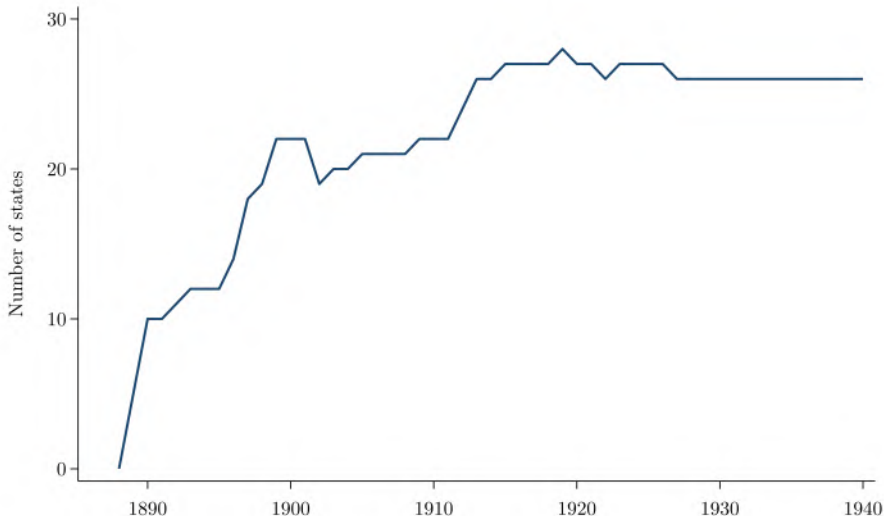
Avg. Statutory Minimum and Maximum Prison Sentences



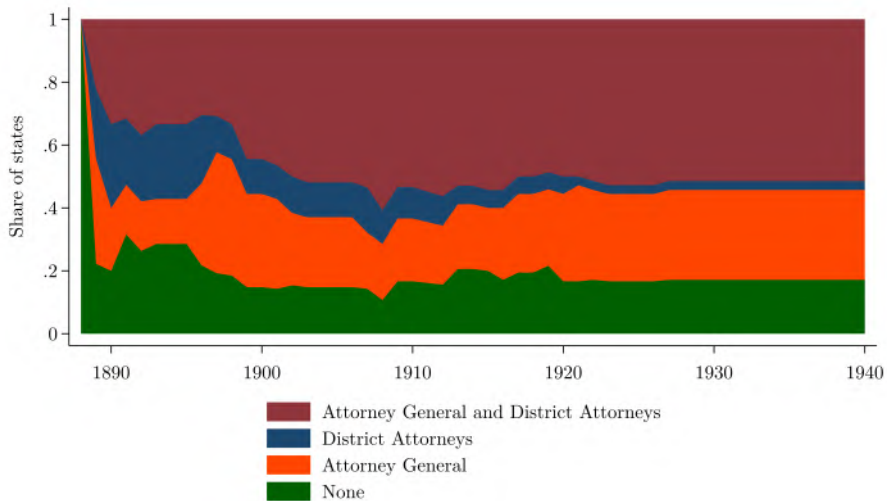
Number of States Making Each Day of Violation a Separate Offense Under State Antitrust Laws



Number of States Allowing Corporate Charters to Be Revoked to Punish Antitrust Violations



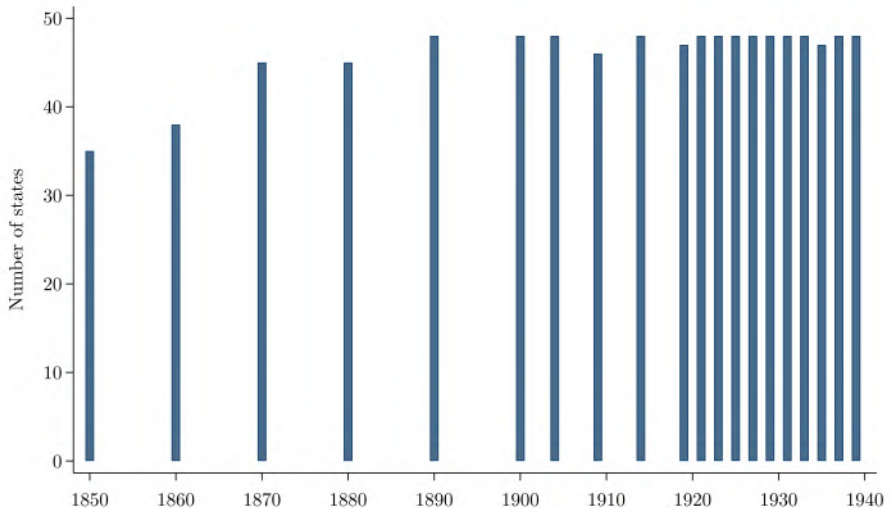
Share of States Designating Enforcement, by Office



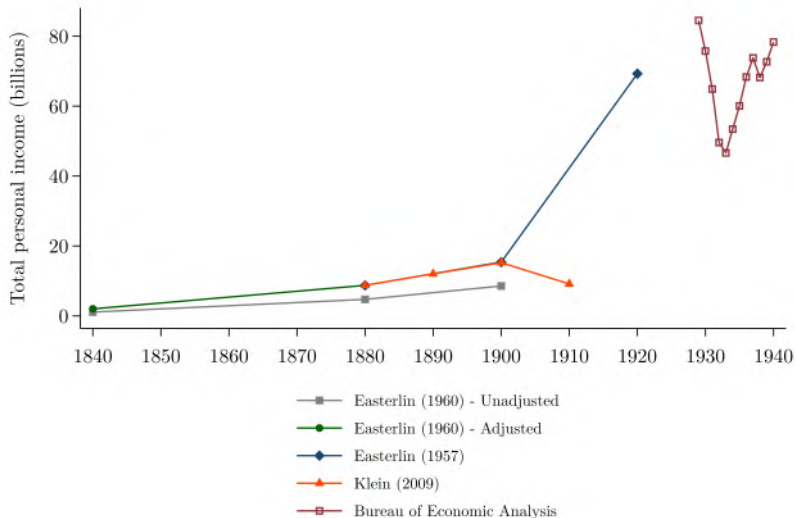
Examples of State Antitrust Cases with Interstate Scope

- Hammond Packing Co. v. Arkansas, 212 U.S. 322 (1909):
The U.S. Supreme Court upheld Arkansas's power to condition an Illinois corporation's right to do business in Arkansas on not belonging to a price-fixing conspiracy *anywhere* (the conspiracy was alleged to have occurred in Illinois)
- International Harvester Co. v. Missouri, 234 U.S. 199 (1914):
The U.S. Supreme Court upheld Missouri's antitrust action against an out-of-state corporation that had formed a monopoly affecting local markets, concluding that the corporation's presence and conduct within the state subjected it to Missouri's jurisdiction

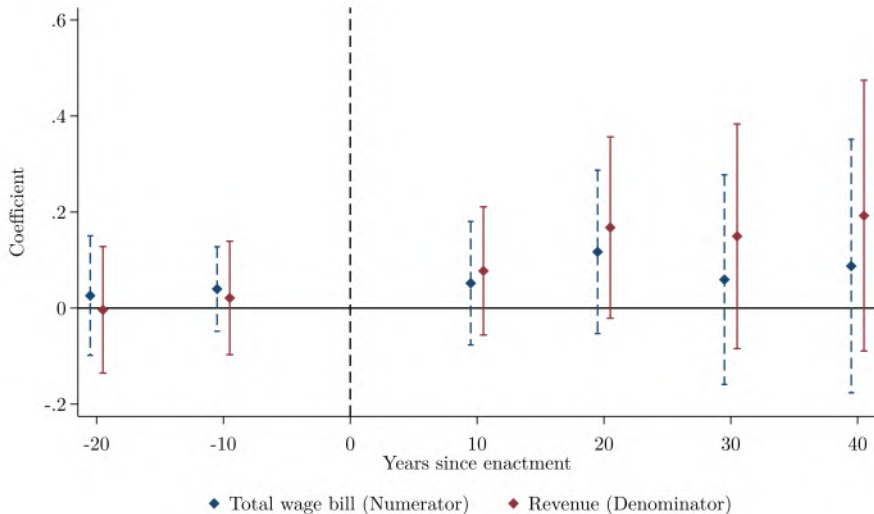
State-by-Industry Census of Manufactures Data Coverage



Personal Income Series



Event Study Results: Labor Share Components

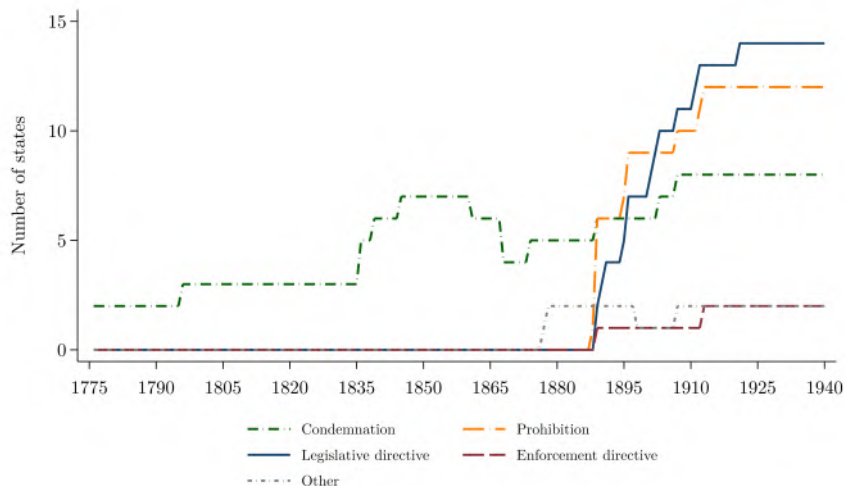


Antitrust Law Index Components

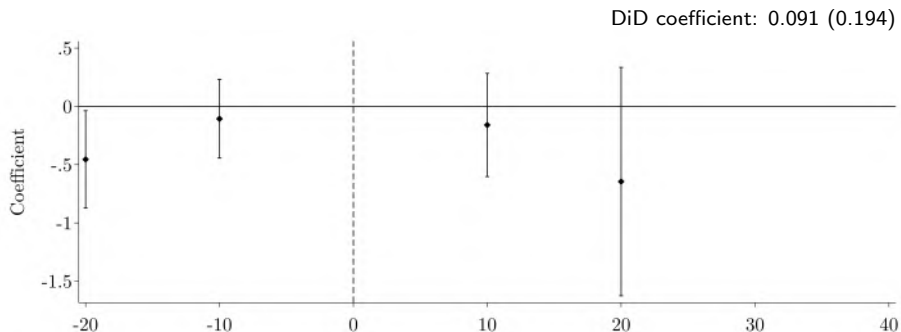
	Variable	First Principal Component	First Principal Component Squared
1	Violating the statute is a crime ¹	0.298	0.089
2	Violating the statute is a tort ¹	0.253	0.064
3	Minimum prison sentence for violating the statute ²	0.148	0.022
4	Maximum prison sentence for violating the statute ²	0.225	0.050
5	Minimum fine for violating the statute ²	0.201	0.040
6	Maximum fine for violating the statute ²	0.142	0.020
7	Level of damages that can be sought for violations ²	0.231	0.053
8	The statute outlaws horizontal price fixing for goods ¹	0.291	0.085
9	The statute outlaws horizontal price fixing for services ¹	0.046	0.002
10	The statute outlaws horizontal output restriction ¹	0.275	0.075
11	The statute outlaws trustee control of corporations ¹	0.223	0.050
12	The statute outlaws restraints of trade ¹	0.278	0.077
13	The statute outlaws monopolization ¹	0.186	0.035
14	The statute outlaws anticompetitive stock purchases ¹	0.148	0.022
15	The statute outlaws refusals to deal ¹	0.167	0.028
16	The statute allows the state to revoke the charters of offending corporations ¹	0.275	0.076
17	The statute allows both a fine and a prison sentence to be imposed ²	0.297	0.088
18	How the statute assigns enforcement responsibilities ²	0.291	0.085
19	The statute stipulates that each day of violation is a separate offense ¹	0.198	0.039
Total			1.000

Notes: "1" denotes an indicator variable and "2" denotes a continuous or ordinal variable.

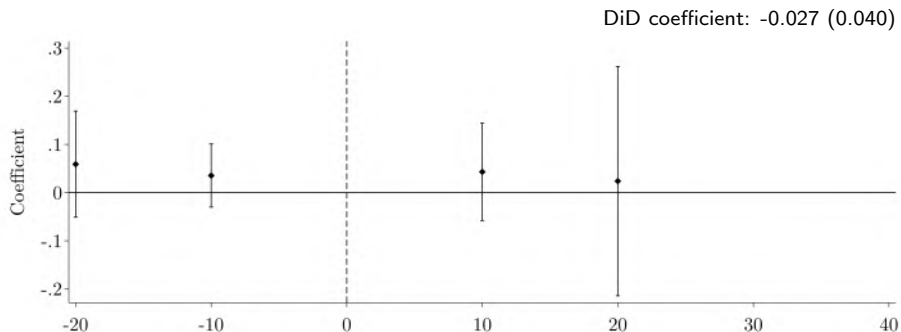
Contents of Anti-Monopoly Provisions in State Constitutions, 1776–1940



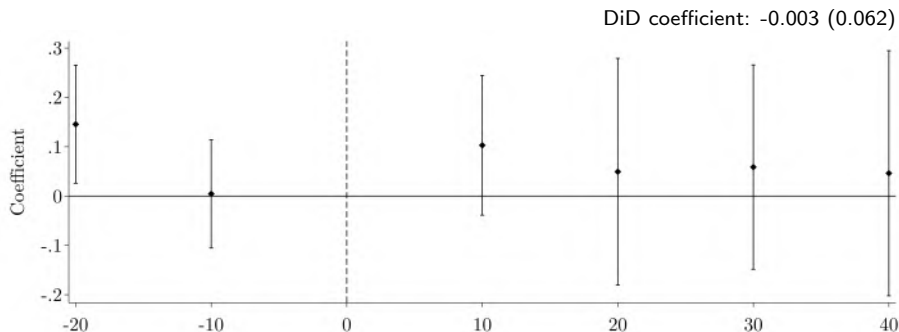
Log Residual Value-Added TFP with Capital



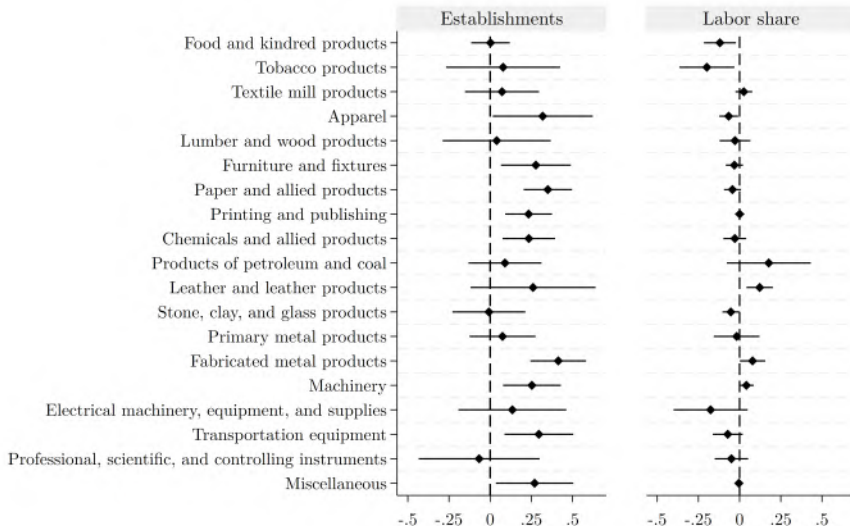
Log Levinsohn and Petrin (2003) TFP with Capital



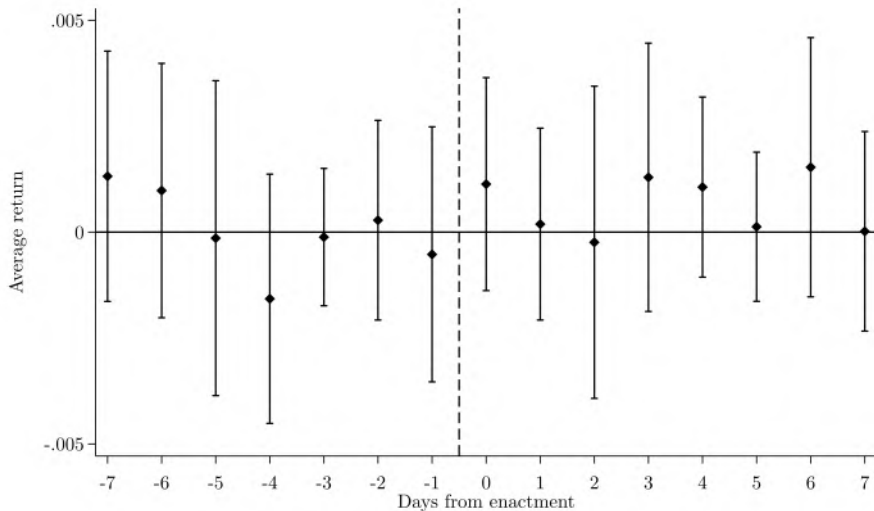
Log Levinsohn and Petrin (2003) TFP with Establishments



Manufacturing Results by Industry Category



Average Daily Stock Returns Around Enactment

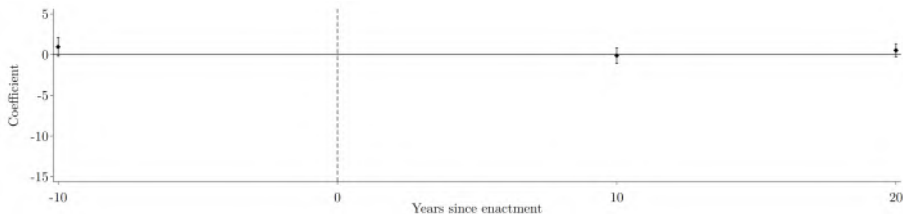


Event Study Results: Number of Trust Incorporations

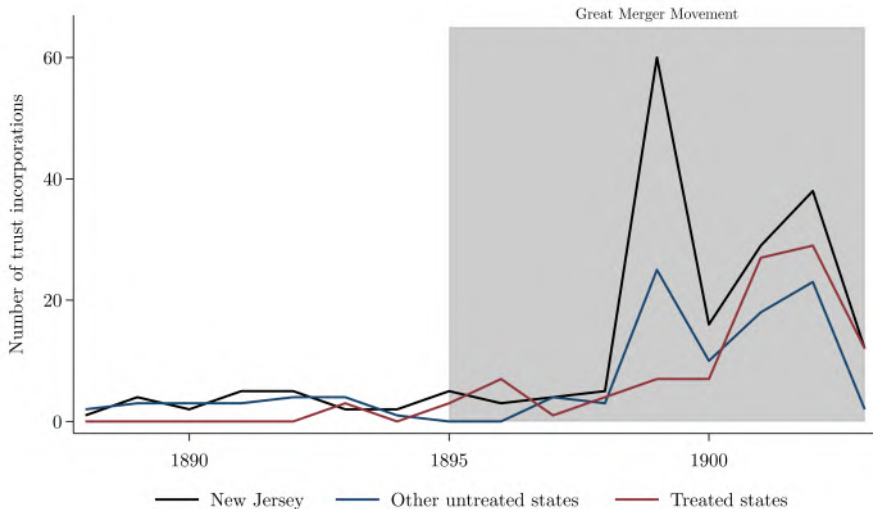
All states



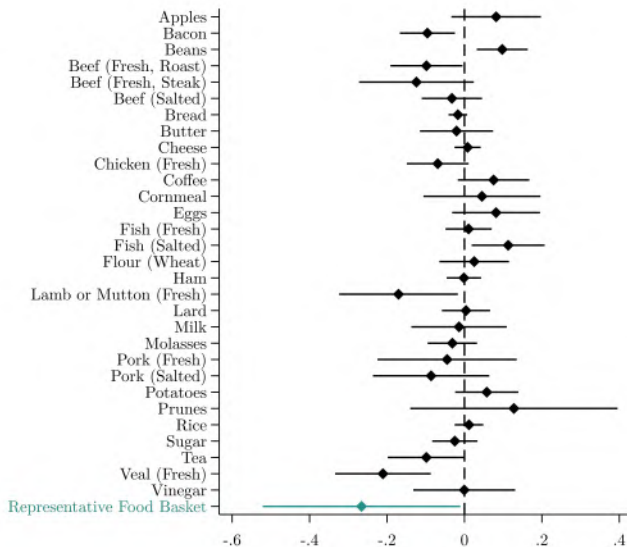
All states except New Jersey



The Great Merger Movement



Effect of Enactment on Food Retail Prices



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